

An introduction to elliptic curve cryptography

- Elliptic curves are examples of commutative groups of "geometric origin."
- Elliptic curves $\neq$ ellipses. Historically, elliptic curves first arose in the computation of so-called "elliptic integrals" related to the problem of finding the arc length of an ellipse.
→ A. Rice, E. Brown: "Why Ellipses Are Not Elliptic Curves."
- Elliptic curves play a colossal role in modern number theory.

Some Key words:

— The "modularity theorem" for "semistable" elliptic curves over $\mathbb{Q}$ was proved by A. Wiles (1995).

This was known to imply Fermat's last theorem

(thanks to previous work of Frey, Serre, and Ribet):

For any $n \geq 3$, there are no positive integers

$$x, y, z \text{ with } x^n + y^n = z^n.$$

— The Birch and Swinnerton-Dyer conjecture (BSD)

predicts a relationship between algebraic properties of the group structure of an elliptic curve over a "number field" (e.g. $\mathbb{Q}$) and analytic properties of an associated function ("Hasse-Weil L-function").

The BSD conjecture is widely regarded as one of the most important and challenging open problems in contemporary mathematics. It is included on the list of seven "Millennium Prize Problems" from 2000.

- Elliptic curves play a major role in modern cryptography.

One reason: while the group operations of elliptic curves can be computed efficiently, no subexponential algorithm for computing discrete logarithms in general elliptic curves over $\mathbb{Z}_p$ is known.

As a consequence, compared to the groups $\mathbb{Z}_p^*$, cryptosystems based on the discrete logarithm problem in elliptic curves require much smaller key sizes for comparable levels of security. (E.g. a key size of 224 bits is viewed as comparable to a 2048 bit RSA key.)

Let F be a "field", e.g. $\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}_p$ (p prime).
 rational, real, or complex numbers

(Fields allow us to add and multiply elements according to the usual rules. They also allow us to divide by non-zero elements)

For technical reasons, we will always assume that

$$1+1 \neq 0 \neq 1+1+1$$

in F ; otherwise, we would have to work harder in the following. Examples of such fields F :

$$\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}_p \text{ (prime, } \underline{p \neq 2, 3})$$

Defⁿ Let $a, b \in F$ with $4a^3 + 27b^2 \neq 0$ in F .

An elliptic curve over F is given by an equation

$$y^2 = x^3 + ax + b, \quad (*)$$

The set of points of the elliptic curve

consists of all those pairs $(x, y) \in \mathbb{F} \times \mathbb{F}$ which satisfy $(*)$ together with another point O , called the point at infinity.

Remark: The condition $4a^3 + 27b^2 \neq 0$ is included to ensure that the curve is "non-singular".

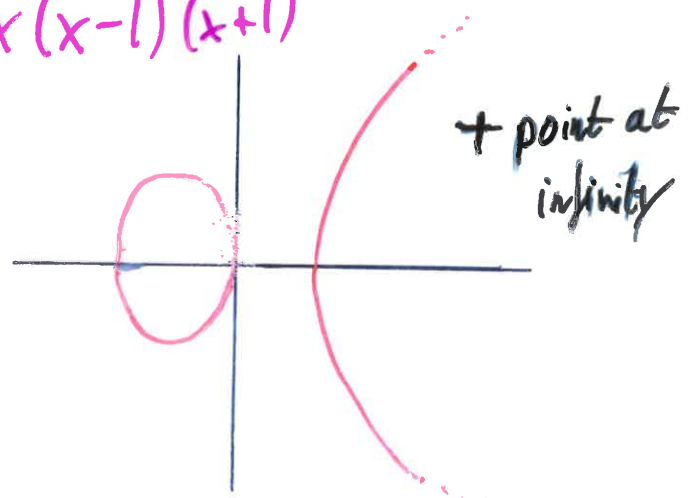
For $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$, this condition is equivalent to the RHS, $x^3 + ax + b$, of $(*)$ having no repeated roots.

Ex: $\mathbb{F} = \mathbb{R}$, $y^2 = x^3 - x = x(x-1)(x+1)$

Some finite points on the curve:

$$(x, y) = (-1, 0), (0, 0), (1, 1),$$

$$\left(-\frac{1}{2}, \pm\sqrt{\frac{3}{8}}\right), \dots$$



The group structure on an elliptic curve

① Geometric approach ("chord-tangent process")

(We'll state this for $F = \mathbb{Q}$ or $F = \mathbb{R}$ but it actually works in all cases.)

Let E be the set of points of an elliptic curve. Let P and Q be points in E .

We define $-P$ and $P+Q$ according to the following rules:

- If $P = \mathcal{O}$ (= point at infinity), then we define $-P := \mathcal{O}$ and $P+Q := Q$.

If $Q = \mathcal{O}$, we define $P+Q := P$.

(Hence, $\mathcal{O}$ acts as an identity element on E w.r.t. $+$.)

Henceforth, suppose that $P \neq \mathcal{O} \neq Q$.

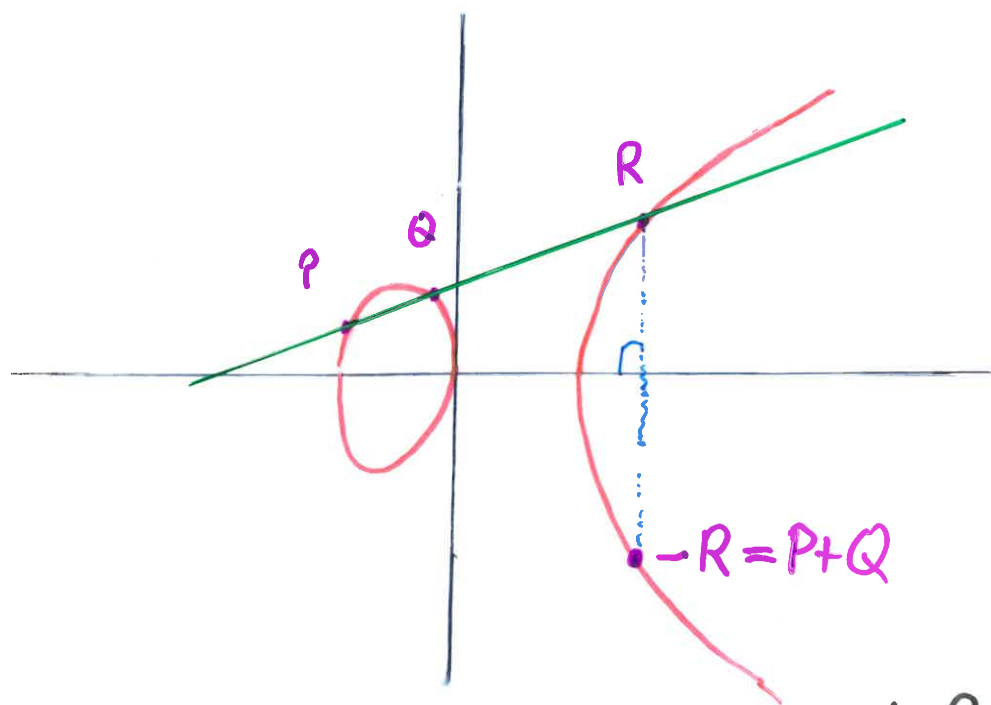
- $-P$ is the point obtained by reflecting P along the x -axis.

Algebraically: $-(x, y) = (x, -y)$.

Note that $(x, y) \in E$ iff $(x, -y) \in E$ due to (*).

- Suppose that P and Q have different x -coordinates.

One can show: the line $\overline{PQ}$ generally intersects E in another point R . We define $P+Q := -R$.



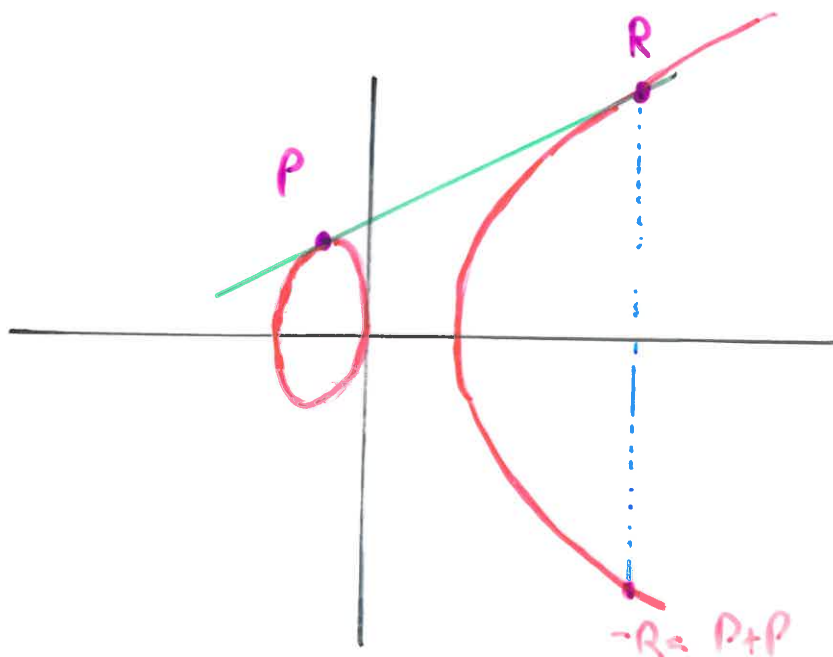
(Exception: if $\overline{PQ}$ is tangent to the curve at P , we take $R = P$; if it is tangent at Q , we take $R = Q$.)

- Suppose that P and Q have the same x -coordinate but $P \neq Q$.

Then $Q = -P$ (and $-Q = P$) and we define

$$P + Q = O.$$

- If $P = Q$, then one can show that, assuming that $P \neq -Q = -P$ (by the preceding step), the tangent line to the curve at P intersects the curve in another point R and we set $P + P := -R$. (When the tangent line only meets the curve in P , we take $R = P$ and so $P + P = -P$.)



Thm $(E, +)$ is an abelian group.
 (Recall: "abelian" and "commutative" are synonymous for groups)

set of points $\nearrow$ $\nearrow$ as defined above

The identity element is the point at infinity $\mathcal{O}$.

The inverse of a point P is $-P$ as defined above.

When people refer to the group law or group structure on an elliptic curve, they mean the operation defined above on the set of points of the curve.

(2) Algebraic approach

By going through the definition of $+$ on E , we can express the group structure in terms of coordinates.

let $P = (x_1, y_1)$ and $Q = (x_2, y_2)$ be
points on the elliptic curve

$$y^2 = x^3 + ax + b,$$

where $P \neq O \neq Q$ and $P \neq -Q$.

Define

$$\lambda := \begin{cases} \frac{y_2 - y_1}{x_2 - x_1}, & \text{if } P \neq Q \\ \frac{3x_1^2 + a}{2y_1}, & \text{if } P = Q. \end{cases}$$

Then $P + Q = (x_3, y_3)$ for

$$x_3 = \lambda^2 - x_1 - x_2 \quad \text{and}$$

$$y_3 = \lambda(x_1 - x_3) - y_1,$$

For proofs, see e.g. Smart, § 3.1.

Summary: For any elliptic curve

$$y^2 = x^3 + ax + b$$

over a field $\mathbb{F}$ in which $2 \neq 0 \neq 3$, we obtained a commutative group structure on the set E of points of the curve. (The point at infinity is the identity element.)

Q: How do elliptic curves look like... as geometric objects or as groups?

Fact (for topologists):

Every elliptic curve over $\mathbb{C}$ is (or rather: its sets of points endowed with the natural topology is) homeomorphic to the torus $S^1 \times S^1 \approx$ . "doughnut"

Elliptic curves over $\mathbb{Z}_p$ (p prime, $p \neq 2, 3$)

(Similarly: over finite fields)

Let E be set of points of the elliptic curve

$$y^2 = x^3 + ax + b$$

over $\mathbb{Z}_p$. Then E is a finite abelian group.

What is its order?

Hasse's theorem: $|E - p - 1| \leq 2\sqrt{p}$

(So the no. of points is close to $p+1$, the no. of points on the "projective line" over $\mathbb{Z}_p$.)

Important: The group E is not necessarily cyclic.

However, it is known that it is always

"made" out of two cyclic groups ($\rightarrow$ direct product).

For cryptographic applications, one works with a large cyclic subgroup of E .

Further details: $\rightarrow$ Koblitz: "A Course in Number Theory and Cryptography"